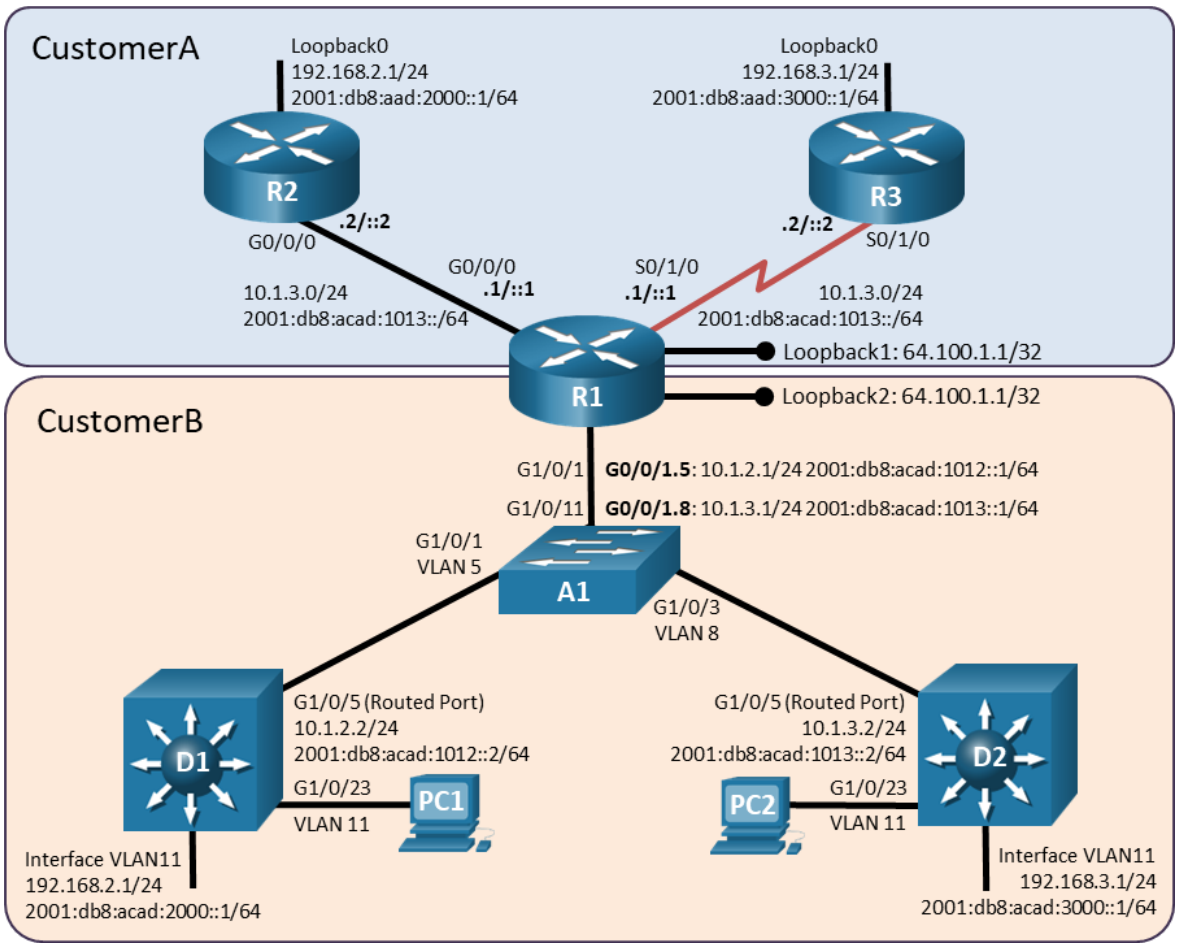
Lab - Implement VRF-Lite

Topology



Addressing Table

Device	Interface	IPv4 Address	IPv6 Address	IPv6 Link-Local
R1	G0/0/0	10.1.2.1/24	2001:db8:acad:1012::1/64	fe80::1:1
	G0/0/1.5	10.1.2.1/24	2001:db8:acad:1012::1/64	fe80::1:2
	G0/0/1.8	10.1.3.1/24	2001:db8:acad:1013::1/64	fe80::1:4
	S0/1/0	10.1.3.1/25	2001:db8:acad:1013::1/64	fe80::1:2
R2	G0/0/0	10.2.3.2/24	2001:db8:acad:1023::2/64	fe80::2:1
	Loopback0	192.168.2.1/24	2001:db8:acad:2000::1/64	fe80::2:2
R3	S0/1/0	10.1.3.3/25	2001:db8:acad:1013::3/64	fe80::3:1

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Device	Interface	IPv4 Address	IPv6 Address	IPv6 Link-Local
	Loopback0	192.168.3.1/27	2001:db8:acad:3000::1/64	fe80::3:2
D1	G1/0/5	10.1.2.2/24	2001:db8:acad:1012::2/64	fe80::d1:1
	VLAN 11	192.168.2.1/24	2001:db8:acad:2000::2/64	fe80::d1:2
D2	G1/0/5	10.1.3.2/24	2001:db8:acad:1013::2/64	fe80::d2:1
	VLAN 11	192.168.3.1/24	2001:db8:acad:3000::1/64	fe80::d2:2

Objectives

Part 1: Build the Network and Configure Basic Device Settings

Part 2: Configure and Verify VRF and Interface Addressing

Part 3: Configure and Verify Dynamic Routing for Reachability Inside Each VRF

Background / Scenario

On a traditional router, all interfaces participate in a single, global routing table. That means every route learned or configured is visible to every interface, and any overlapping IP address space would immediately conflict. Service providers and large enterprises often need to virtualize a router so that it can act like multiple independent routers at once.

Virtual Routing and Forwarding (VRF) solves this by creating separate routing tables on the same device. Each table is an isolated routing instance with its own interfaces and routes. VRF-Lite delivers this capability without Multiprotocol Label Switching (MPLS) and is intended for local segmentation on a single router or a small network.

With VRF-Lite, you place interfaces into specific VRFs. Each VRF maintains its own routes, so identical prefixes can exist at the same time without interference. The router forwards based on the VRF tied to the incoming interface, which keeps customer or department traffic isolated. Traffic does not cross between VRFs by default, which prevents leakage. Reachability inside each VRF uses familiar methods such as static routes or dynamic routing protocols configured per VRF. Inter-VRF communication, if needed, must be configured explicitly.

In this lab, you will work on R1, acting as a service provider router that supports two customers using the same addressing scheme. Your task is to deploy VRF-Lite and static routing to ensure full reachability within each customer's network.

Note: This lab is an exercise in developing, deploying, and verifying VRF-Lite, and does not reflect networking best practices.

Note: The routers and switches used with CCNP hands-on labs are Cisco 8200L, Cisco 9300L, and Cisco 9200L with Cisco IOS XE Release 17.12.05 (universalk9 image). Other routers, switches, and Cisco IOS versions can be used. Depending on the model and Cisco IOS version, the commands available and the output produced might vary from what is shown in the labs

Note: Ensure that the routers and switches have been erased and have no startup configurations. If you are unsure contact your instructor.

Note: The PCs used in this lab do not require addressing. They are needed to enable interface VLAN 11.

Required Resources

- 3 Routers with a compatible serial interface (Cisco 8200L with Cisco IOS XE Release 17.12.05c universal image or comparable)

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- 2 Switches (Cisco 9300L with Cisco IOS XE release 17.12.05 universal image or comparable)
- 1 Switch (Cisco 9200L with Cisco IOS XE release 17.12.05 universal image or comparable)
- 2 PCs (Windows with a terminal emulation program, such as Tera Term)
- Console cables to configure the Cisco IOS devices via the console ports
- Ethernet and serial cables as shown in the topology

Part 1: Build the Network and Configure Basic Device Settings

In Part 1, you will set up the network topology and configure basic settings on all devices.

Step 1: Cable the network as shown in the topology.

Attach the devices as shown in the topology diagram, and cable as necessary.

Step 2: Configure basic settings for each device.

a. Console into each device, enter global configuration mode, and apply the basic settings. A command list for each device using the following startup configurations.

Router R1

```
enable
configure terminal
hostname R1
no ip domain lookup
ipv6 unicast-routing
banner motd # R1, Implement VRF-Lite #
line con 0
 exec-timeout 0 0
 logging synchronous
 exit
line vty 0 4
 privilege level 15
 password cisco123
 exec-timeout 0 0
 logging synchronous
 login
 exit
```

Router R2

```
enable
configure terminal
hostname R2
no ip domain lookup
ipv6 unicast-routing
banner motd # R2, Implement VRF-Lite #
line con 0
 exec-timeout 0 0
 logging synchronous
```

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```
 exit
line vty 0 4
 privilege level 15
 password cisco123
 exec-timeout 0 0
 logging synchronous
 login
 exit
interface g0/0/0
 ip address 10.1.2.2 255.255.255.0
 ipv6 address fe80::2:1 link-local
 ipv6 address 2001:db8:acad:1012::2/64
 ospfv3 100 ipv6 area 0
 ospfv3 100 ipv4 area 0
 no shutdown
 exit
interface loopback 0
 ip address 192.168.2.1 255.255.255.0
 ipv6 address fe80::2:2 link-local
 ipv6 address 2001:db8:acad:2000::1/64
 ospfv3 network point-to-point
 ospfv3 100 ipv4 area 0
 ospfv3 100 ipv6 area 0
 no shutdown
 exit
router ospfv3 100
 router-id 2.2.2.2
 address-family ipv4 unicast
 exit-address-family
 address-family ipv6 unicast
 exit-address-family
```

Router R3

```
enable
configure terminal
hostname R3
no ip domain lookup
ipv6 unicast-routing
banner motd # R3, Implement VRF-Lite #
line con 0
 exec-timeout 0 0
 logging synchronous
 exit
line vty 0 4
 privilege level 15
 password cisco123
```

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```
 exec-timeout 0 0
 logging synchronous
 login
 exit
interface loopback 0
 ip address 192.168.3.1 255.255.255.0
 ipv6 address fe80::3:2 link-local
 ipv6 address 2001:db8:acad:3000::1/64
 ospfv3 network point-to-point
 ospfv3 100 ipv4 area 0
 ospfv3 100 ipv6 area 0
 no shutdown
 exit
interface s0/1/0
 ip address 10.1.3.2 255.255.255.0
 ipv6 address fe80::3:1 link-local
 ipv6 address 2001:db8:acad:1013::2/64
 ospfv3 100 ipv4 area 0
 ospfv3 100 ipv6 area 0
 no shutdown
 exit
router ospfv3 100
 router-id 3.3.3.3
 address-family ipv4 unicast
 exit-address-family
 address-family ipv6 unicast
 exit-address-family
```

Switch D1

```
enable
configure terminal
hostname D1
no ip domain lookup
ip routing
ipv6 unicast-routing
banner motd # D1, Implement VRF-Lite #
line con 0
 exec-timeout 0 0
 logging synchronous
 exit
line vty 0 4
 privilege level 15
 password cisco123
 exec-timeout 0 0
 logging synchronous
 login
```

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```
 exit
interface range g1/0/1-24, g1/1/1-4, g0/0
 shutdown
 exit
interface g1/0/5
 no switchport
 ip address 10.1.2.2 255.255.255.0
 ipv6 address fe80::d1:1 link-local
 ipv6 address 2001:db8:acad:1012::2/64
 ospfv3 200 ipv4 area 0
 ospfv3 200 ipv6 area 0
 no shutdown
 exit
vlan 11
 name LOCAL_VLAN
 exit
interface vlan 11
 ip address 192.168.2.1 255.255.255.0
 ipv6 address fe80::d1:2 link-local
 ipv6 address 2001:db8:acad:2000::1/64
 ospfv3 200 ipv4 area 0
 ospfv3 200 ipv6 area 0
 no shutdown
 exit
interface g1/0/23
 switchport mode access
 switchport access vlan 11
 no shutdown
 exit
router ospfv3 200
 router-id 4.4.4.4
 address-family ipv4 unicast
 exit-address-family
 address-family ipv6 unicast
 exit-address-family
```

Switch D2

```
enable
configure terminal
hostname D2
no ip domain lookup
ip routing
ipv6 unicast-routing
banner motd # D2, Implement VRF-Lite #
line con 0
 exec-timeout 0 0
```

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```
 logging synchronous
 exit
line vty 0 4
 privilege level 15
 password cisco123
 exec-timeout 0 0
 logging synchronous
 login
 exit
interface range g1/0/1-24, g1/1/1-4, g0/0
 shutdown
 exit
interface g1/0/5
 no switchport
 ip address 10.1.3.2 255.255.255.0
 ipv6 address fe80::d2:1 link-local
 ipv6 address 2001:db8:acad:1013::2/64
 ospfv3 200 ipv4 area 0
 ospfv3 200 ipv6 area 0
 no shutdown
 exit
vlan 11
 name LOCAL_VLAN
 exit
interface vlan 11
 ip address 192.168.3.1 255.255.255.0
 ipv6 address fe80::d2:2 link-local
 ipv6 address 2001:db8:acad:3000::1/64
 ospfv3 200 ipv4 area 0
 ospfv3 200 ipv6 area 0
 no shutdown
 exit
interface g1/0/23
 switchport mode access
 switchport access vlan 11
 no shutdown
 exit
router ospfv3 200
 router-id 5.5.5.5
 address-family ipv4 unicast
 exit-address-family
 address-family ipv6 unicast
 exit-address-family
```

Switch A1

```
enable
```

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```
configure terminal
hostname A1
no ip domain lookup
banner motd # A1, Implement VRF-Lite #
line con 0
 exec-timeout 0 0
 logging synchronous
 exit
line vty 0 4
 privilege level 15
 password cisco123
 exec-timeout 0 0
 logging synchronous
 login
 exit
interface range g1/0/1-24, g1/1/1-4, g0/0
 shutdown
 exit
vlan 5
 name D1
 exit
vlan 8
 name D2
 exit
interface g1/0/11
 switchport mode trunk
 switchport nonegotiate
 no shutdown
 exit
interface g1/0/1
 switchport mode access
 switchport access vlan 5
 no shutdown
 exit
interface g1/0/3
 switchport mode access
 switchport access vlan 8
 no shutdown
```

b. Set the clock on each router to UTC time.

c. Save the running configuration to startup-config.

Part 2: Configure and Verify VRF and Interface Addressing

In Part 2, you will configure and verify VRF-Lite on R1. The other devices, R2, R3, D1, D2, and A1 require no additional configuration. Once again, the configuration being used here is not meant to represent best practice, but to assess your ability to complete the required configurations.

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Step 1: On R1, create the required VRFs.

a. Create the CustomerA and CustomerB VRFs, and initialize them for both IPv4 and IPv6. The VRF names are case sensitive.

```
R1(config)# vrf definition CustomerA
R1(config-vrf)# address-family ipv4
R1(config-vrf-af)# address-family ipv6
R1(config-vrf-af)# exit
R1(config-vrf)# vrf definition CustomerB
R1(config-vrf)# address-family ipv4
R1(config-vrf-af)# address-family ipv6
R1(config-vrf-af)# exit
```

b. Configure interfaces G0/0/0 and S0/1/0 for the CustomerA network.

```
R1(config)# interface g0/0/0
R1(config-if)# vrf forwarding CustomerA
R1(config-if)# ip address 10.1.2.1 255.255.255.0
R1(config-if)# ipv6 address fe80::1:1 link-local
R1(config-if)# ipv6 address 2001:db8:acad:1012::1/64
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)# interface s0/1/0
R1(config-if)# vrf forwarding CustomerA
R1(config-if)# ip address 10.1.3.1 255.255.255.0
R1(config-if)# ipv6 address fe80::1:4 link-local
R1(config-if)# ipv6 address 2001:db8:acad:1013::1/64
R1(config-if)# no shutdown
R1(config-if)# exit
```

c. Configure R1 interface G0/0/1 to support the CustomerB networks. G0/0/1 will be performing inter-VLAN routing between VLANs 5 and 8.

```
R1(config)# interface g0/0/1
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)# interface g0/0/1.5
R1(config-subif)# encapsulation dot1q 5
R1(config-subif)# vrf forwarding CustomerB
R1(config-subif)# ip address 10.1.2.1 255.255.255.0
R1(config-subif)# ipv6 address fe80::1:2 link-local
R1(config-subif)# ipv6 address 2001:db8:acad:1012::1/64
R1(config-subif)# exit
R1(config)# interface g0/0/1.8
R1(config-subif)# encapsulation dot1q 8
R1(config-subif)# vrf forwarding CustomerB
R1(config-subif)# ip address 10.1.3.1 255.255.255.0
R1(config-subif)# ipv6 address fe80::1:3 link-local
R1(config-subif)# ipv6 address 2001:db8:acad:1013::1/64
```

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```
R1(config-subif)# exit
```

d. Configure loopback interfaces on R1 to simulate access to external networks.

```
R1(config)# interface Loopback1
R1(config-if)# description simulated access to external networks
R1(config-if)# vrf forwarding CustomerA
R1(config-if)# ip address 64.100.1.1 255.255.255.252
R1(config-if)# ipv6 address 2001:db8:acad:64:100:1::1/64
R1(config-if)# interface Loopback2
R1(config-if)# description simulated access to external networks
R1(config-if)# vrf forwarding CustomerB
R1(config-if)# ip address 64.100.1.1 255.255.255.252
R1(config-if)# ipv6 address 2001:db8:acad:64:100:1::1/64
```

Step 2: Verify the VRF-Lite configuration.

a. Verify the routing and forwarding information.

```
R1# show ip vrf
  Name                              Default RD            Interfaces
  CustomerA                         <not set>             Gi0/0/0
                                                          Se0/1/0
                                                          Lo1
  CustomerB                         <not set>             Gi0/0/1.5
                                                          Gi0/0/1.8
                                                          Lo2
```

b. Verify the interface assignments using the **show ip vrf interfaces** command.

```
R1# show ip vrf interfaces
Interface              IP-Address      VRF                              Protocol
Gi0/0/0                10.1.2.1        CustomerA                        up
Se0/1/0                10.1.3.1        CustomerA                        up
Lo1                    64.100.1.1      CustomerA                        up
Gi0/0/1.5              10.1.2.1        CustomerB                        up
Gi0/0/1.8              10.1.3.1        CustomerB                        up
Lo2                    64.100.1.1      CustomerB                        up
```

c. Verify the VRF routing tables with the **show ip route vrf** *vrf_name* and **show ipv6 route vrf** *vrf_name* command.

```
R1# show ip route vrf CustomerA | begin Gateway
Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C        10.1.2.0/24 is directly connected, GigabitEthernet0/0/0
L        10.1.2.1/32 is directly connected, GigabitEthernet0/0/0
C        10.1.3.0/24 is directly connected, Serial0/1/0
L        10.1.3.1/32 is directly connected, Serial0/1/0
      64.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        64.100.1.0/30 is directly connected, Loopback1
L        64.100.1.1/32 is directly connected, Loopback1
```

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```
R1# show ipv6 route vrf CustomerB
IPv6 Routing Table - CustomerB - 5 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route
<output omitted>
C   2001:DB8:ACAD:64::/64 [0/0]
     via Loopback2, directly connected
L   2001:DB8:ACAD:64:100:1:0:1/128 [0/0]
     via Loopback2, receive
C   2001:DB8:ACAD:1012::/64 [0/0]
     via GigabitEthernet0/0/1.5, directly connected
L   2001:DB8:ACAD:1012::1/128 [0/0]
     via GigabitEthernet0/0/1.5, receive
C   2001:DB8:ACAD:1013::/64 [0/0]
     via GigabitEthernet0/0/1.8, directly connected
L   2001:DB8:ACAD:1013::1/128 [0/0]
     via GigabitEthernet0/0/1.8, receive
L   FF00::/8 [0/0]
     via Null0, receive
```

d. Verify next-hop reachability within each vrf with the **ping vrf** *vrf_name* **address** command.

```
R1# ping vrf CustomerA 10.1.2.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.1.2.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 1/1/1 ms
R1# ping vrf CustomerA 2001:db8:acad:1012::2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:ACAD:1012::2, timeout is 2
seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/2/10 ms
R1#  ping vrf CustomerA 10.1.3.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.1.3.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 2/2/3 ms
R1# ping vrf CustomerA 2001:db8:acad:1013::2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:ACAD:1013::2, timeout is 2
seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 2/2/3 ms
```

Part 3: Configure and Verify Dynamic Routing for Reachability Inside Each VRF

In Part 3, you will configure OSFP so that all networks are reachable within their respective VRFs. At the end of this part, R1 should be able to successfully source a ping from interface loopback0 to R3 interface loopback0, and D1 should be able to successfully source a ping from interface VLAN 11 to D2 interface VLAN

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11. Furthermore, you will configure a default route to access an external network simulated the loopback interfaces. Once again, the way these networks are being implemented is not meant to represent best practice, but to assess your ability to complete the required configurations.

Step 1: Verify that distant networks are not reachable within each VRF.

In this step, you will check to make sure that distant networks are not reachable from R1 within each VRF.

a. On R1, issue the commands **ping vrf CustomerA 192.168.2.1** and **ping vrf CustomerA 192.168.3.1**. Neither should succeed.

```
R1# ping vrf CustomerA 192.168.2.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.2.1, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)
R1# ping vrf CustomerA 192.168.3.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.3.1, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)
```

b. On R1, issue the commands **ping vrf CustomerA 2001:db8:acad:2000::1** and **ping vrf CustomerA 2001:db8:acad:3000::1**. Neither should succeed.

```
R1# ping vrf CustomerA 2001:db8:acad:2000::1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:ACAD:2000::1, timeout is 2
seconds:

% No valid route for destination
Success rate is 0 percent (0/1)
R1# ping vrf CustomerA 2001:db8:acad:3000::1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:ACAD:3000::1, timeout is 2
seconds:

% No valid route for destination
Success rate is 0 percent (0/1)
```

c. On R1, issue the commands **ping vrf CustomerB 192.168.2.1** and **ping vrf CustomerB 192.168.3.1**. Neither should succeed.

```
R1# ping vrf CustomerB 192.168.2.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.2.1, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)
R1# ping vrf CustomerB 192.168.3.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.3.1, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)
```

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d. On R1, issue the commands **ping vrf CustomerB 2001:db8:acad:2000::1** and **ping vrf CustomerB 2001:db8:acad:3000::1**. Neither should succeed.

```
R1# ping vrf CustomerB 2001:db8:acad:2000::1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:ACAD:2000::1, timeout is 2 seconds:

% No valid route for destination
Success rate is 0 percent (0/1)

R1# ping vrf CustomerB 2001:db8:acad:3000::1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:ACAD:3000::1, timeout is 2 seconds:

% No valid route for destination
Success rate is 0 percent (0/1)
```

Step 2: Configure OSFP at R1 for each VRF.

In this step, you will configure R1 so that it can reach distant networks in each VRF. The neighbor systems (D1, D2, R2, and R3) have OSPF routes already configured, so as soon as you correctly install these dynamic routes, there will be full reachability within each VRF.

a. On R1, configure OSPF routes for the distant networks in the CustomerA VRF using OSPFv3.

```
R1(config)# interface GigabitEthernet0/0/0
R1(config-if)# ospfv3 100 ipv4 area 0
R1(config-if)# ospfv3 100 ipv6 area 0
R1(config-if)# interface Serial0/1/0
R1(config-if)# ospfv3 100 ipv4 area 0
R1(config-if)# ospfv3 100 ipv6 area 0
R1(config-if)# router ospfv3 100
R1(config-router)# router-id 1.1.1.1
R1(config-router)# address-family ipv4 unicast vrf CustomerA
R1(config-router-af)# capability vrf-lite
R1(config-router-af)# exit-address-family
R1(config-router)# address-family ipv6 unicast vrf CustomerA
R1(config-router-af)# capability vrf-lite
R1(config-router-af)# exit-address-family
R1(config-router)# exit
```

b. Use the example above to correctly configure the OSPF routes for the CustomerB network.

Step 3: Verify OSPF routes.

a. On R1, verify the OSPF routes in routing table for CustomerA.

```
R1# show ip route vrf CustomerA ospfv3 | begin Gateway
Gateway of last resort is not set

O     192.168.2.0/24 [110/2] via 10.1.2.2, 00:23:07, GigabitEthernet0/0/0
O     192.168.3.0/24 [110/51] via 10.1.3.2, 00:23:07, Serial0/1/0
```

b. On R2 and R3, display and verify the OSPF routes in the routing table.

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```
R2# show ip route ospfv3 | begin Gateway
Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
O        10.1.3.0/24 [110/51] via 10.1.2.1, 00:25:47, GigabitEthernet0/0/0
O     192.168.3.0/24 [110/52] via 10.1.2.1, 00:25:43, GigabitEthernet0/0/0

R3# show ip route ospfv3 | begin Gateway
Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
O        10.1.2.0/24 [110/50] via 10.1.3.1, 00:26:16, Serial0/1/0
O     192.168.2.0/24 [110/51] via 10.1.3.1, 00:26:16, Serial0/1/0
```

c. Verify the OSPF routes for CustomerB.

```
R1# show ip route vrf CustomerB ospfv3 | begin Gateway
Gateway of last resort is not set

O     192.168.2.0/24 [110/2] via 10.1.2.2, 00:27:42, GigabitEthernet0/0/1.5
O     192.168.3.0/24 [110/2] via 10.1.3.2, 00:27:42, GigabitEthernet0/0/1.8

D1# show ip route ospfv3 | begin Gateway
Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
O        10.1.3.0/24 [110/2] via 10.1.2.1, 00:28:33, GigabitEthernet1/0/5
O     192.168.3.0/24 [110/3] via 10.1.2.1, 00:28:28, GigabitEthernet1/0/5

D2# show ip route ospfv3 | begin Gateway
Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
O        10.1.2.0/24 [110/2] via 10.1.3.1, 00:29:08, GigabitEthernet1/0/5
O     192.168.2.0/24 [110/3] via 10.1.3.1, 00:29:08, GigabitEthernet1/0/5
```

d. Repeat the above steps for IPv6 OSPF routes.

```
R1# show ipv6 route vrf CustomerA ospf
<output omitted>
O   2001:DB8:ACAD:2000::/64 [110/2]
     via FE80::2:1, GigabitEthernet0/0/0
O   2001:DB8:ACAD:3000::/64 [110/51]
     via FE80::3:1, Serial0/1/0

R2# show ipv6 route ospf
<output omitted>
O   2001:DB8:ACAD:1013::/64 [110/51]
     via FE80::1:1, GigabitEthernet0/0/0
O   2001:DB8:ACAD:3000::/64 [110/52]
     via FE80::1:1, GigabitEthernet0/0/0
```

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```
R3# show ipv6 route ospf
<output omitted>
O   2001:DB8:ACAD:1012::/64 [110/50]
     via FE80::1:4, Serial0/1/0
O   2001:DB8:ACAD:2000::/64 [110/51]
     via FE80::1:4, Serial0/1/0
```

e. Because all the interfaces are participating in the VRF, there are no routes in global routing table.

```
R1# show ip route | begin Gateway
Gateway of last resort is not set


R1# show ipv6 route
IPv6 Routing Table - default - 1 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route
       B - BGP, R - RIP, H - NHRP, HG - NHRP registered
       Hg - NHRP registration summary, HE - NHRP External, I1 - ISIS L1
       I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary, D - EIGRP
       EX - EIGRP external, ND - ND Default, NDp - ND Prefix, DCE -
Destination
       NDr - Redirect, O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1
       OE2 - OSPF ext 2, ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
       a - Application, m - OMP
L   FF00::/8 [0/0]
     via Null0, receive
```

Step 4: Verify full reachability within each VRF.

a. On R2, ping the IPv4 and IPv6 addresses of R3 interface Loopback0 using a source address of R2 interface Loopback0. All pings should be successful.

```
R2# ping 192.168.3.1 source loopback0
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.3.1, timeout is 2 seconds:
Packet sent with a source address of 192.168.2.1
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 2/2/3 ms
R2# ping 2001:db8:acad:3000::1 source loopback0
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:ACAD:3000::1, timeout is 2 seconds:
Packet sent with a source address of 2001:DB8:ACAD:2000::1
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 2/2/2 ms
```

b. On D1, ping the IPv4 and IPv6 addresses of D2 interface VLAN 11 using a source address of D1 interface VLAN 11. All pings should be successful.

```
D1# ping 192.168.3.1 source vlan11


Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.3.1, timeout is 2 seconds:
```

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```
Packet sent with a source address of 192.168.2.1
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/4/9 ms

D1# ping 2001:db8:acad:3000::1 source vlan11

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:ACAD:3000::1, timeout is 2 seconds:
Packet sent with a source address of 2001:DB8:ACAD:2000::1
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/5/17 ms
```

Step 5: Configure and distribute default route for each VRF.

In this step, you will configure default routes on R1 and advertise the default routes into each VRF so that the devices in each VRF can reach distant networks simulated by the loopback interfaces on R1.

a. On R1, configure the default routes for each VRF.

```
R1(config)# ip route vrf CustomerA 0.0.0.0 0.0.0.0 lo 1 64.100.1.2
R1(config)# ipv6 route vrf CustomerA ::/0 lo 1 2001:db8:acad:64:100:1::2
R1(config)# ip route vrf CustomerB 0.0.0.0 0.0.0.0 lo 2 64.100.1.2
R1(config)# ipv6 route vrf CustomerB ::/0 lo 2 2001:db8:acad:64:100:1::2
```

b. Advertise the default routes into the OSPF process 100 and 200 so each VRF member learns the default routes.

```
R1(config)# router ospfv3 100
R1(config-router)# address-family ipv4 unicast vrf CustomerA
R1(config-router-af)# default-information originate
R1(config-router-af)# exit-address-family
R1(config-router)# address-family ipv6 unicast vrf CustomerA
R1(config-router-af)# default-information originate
R1(config-router-af)# exit-address-family
R1(config-router)# router ospfv3 200
R1(config-router)# address-family ipv4 unicast vrf CustomerB
R1(config-router-af)# default-information originate
R1(config-router-af)# exit-address-family
R1(config-router)# address-family ipv6 unicast vrf CustomerB
R1(config-router-af)# default-information originate
R1(config-router-af)# exit-address-family
```

c. Verify that default routes have been advertised with each VRF.

```
R1# show ip route vrf CustomerA | begin Gateway
Gateway of last resort is 64.100.1.2 to network 0.0.0.0

S*    0.0.0.0/0 [1/0] via 64.100.1.2, Loopback1
      10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C        10.1.2.0/24 is directly connected, GigabitEthernet0/0/0
L        10.1.2.1/32 is directly connected, GigabitEthernet0/0/0
C        10.1.3.0/24 is directly connected, Serial0/1/0
L        10.1.3.1/32 is directly connected, Serial0/1/0
```

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```
      64.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        64.100.1.0/30 is directly connected, Loopback1
L        64.100.1.1/32 is directly connected, Loopback1
O     192.168.2.0/24 [110/2] via 10.1.2.2, 00:10:22, GigabitEthernet0/0/0
O     192.168.3.0/24 [110/51] via 10.1.3.2, 00:10:21, Serial0/1/0
```

```
R1# show ipv6 route vrf CustomerA
<output omitted>
S   ::/0 [1/0]
     via 2001:DB8:ACAD:64:100:1:0:2, Loopback1
C   2001:DB8:ACAD:64::/64 [0/0]
     via Loopback2, directly connected
L   2001:DB8:ACAD:64:100:1:0:1/128 [0/0]
     via Loopback2, receive
C   2001:DB8:ACAD:1012::/64 [0/0]
     via GigabitEthernet0/0/1.5, directly connected
L   2001:DB8:ACAD:1012::1/128 [0/0]
     via GigabitEthernet0/0/1.5, receive
C   2001:DB8:ACAD:1013::/64 [0/0]
     via GigabitEthernet0/0/1.8, directly connected
L   2001:DB8:ACAD:1013::1/128 [0/0]
     via GigabitEthernet0/0/1.8, receive
O   2001:DB8:ACAD:2000::/64 [110/2]
     via FE80::D1:1, GigabitEthernet0/0/1.5
O   2001:DB8:ACAD:3000::/64 [110/2]
     via FE80::D2:1, GigabitEthernet0/0/1.8
L   FF00::/8 [0/0]
     via Null0, receive
```

```
R2# show ip route | begin Gateway
Gateway of last resort is 10.1.2.1 to network 0.0.0.0

O*E2  0.0.0.0/0 [110/1] via 10.1.2.1, 00:16:15, GigabitEthernet0/0/0
      10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
C        10.1.2.0/24 is directly connected, GigabitEthernet0/0/0
L        10.1.2.2/32 is directly connected, GigabitEthernet0/0/0
O        10.1.3.0/24 [110/51] via 10.1.2.1, 00:16:15, GigabitEthernet0/0/0
      192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.2.0/24 is directly connected, Loopback0
L        192.168.2.1/32 is directly connected, Loopback0
O     192.168.3.0/24 [110/52] via 10.1.2.1, 00:14:44, GigabitEthernet0/0/0
```

```
R3# show ipv6 route
<output omitted>
OE2 ::/0 [110/1], tag 100
     via FE80::1:4, Serial0/1/0
O   2001:DB8:ACAD:1012::/64 [110/50]
     via FE80::1:4, Serial0/1/0
C   2001:DB8:ACAD:1013::/64 [0/0]
```

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```
     via Serial0/1/0, directly connected
L    2001:DB8:ACAD:1013::2/128 [0/0]
     via Serial0/1/0, receive
O    2001:DB8:ACAD:2000::/64 [110/51]
     via FE80::1:4, Serial0/1/0
C    2001:DB8:ACAD:3000::/64 [0/0]
     via Loopback0, directly connected
L    2001:DB8:ACAD:3000::1/128 [0/0]
     via Loopback0, receive
L    FF00::/8 [0/0]
     via Null0, receive
```

d. Review the routing protocol parameters and statistics for each VRF.

```
R1# show ip protocols vrf CustomerA
*** IP Routing is NSF aware ***

Routing Protocol is "ospfv3 100"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 1.1.1.1
  Autonomous system boundary router
  Number of areas: 1 normal, 0 stub, 0 nssa
  Interfaces (Area 0):
    Serial0/1/0
    GigabitEthernet0/0/0
  Maximum path: 4
  Routing Information Sources:
    Gateway         Distance      Last Update
    3.3.3.3              110      00:07:48
    2.2.2.2              110      00:09:16
  Distance: (default is 110)

R1# show ip protocols vrf CustomerB
*** IP Routing is NSF aware ***

Routing Protocol is "ospfv3 200"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 1.1.1.1
  Autonomous system boundary router
  Number of areas: 1 normal, 0 stub, 0 nssa
  Interfaces (Area 0):
    GigabitEthernet0/0/1.8
    GigabitEthernet0/0/1.5
  Maximum path: 4
  Routing Information Sources:
    Gateway         Distance      Last Update
    5.5.5.5              110      00:02:14
    4.4.4.4              110      00:04:26
```

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```
  Distance: (default is 110)
```

e. Verify that R2, R3, D1 and D2 can ping the simulated access to the external network.

```
R2# ping 64.100.1.1 source lo0
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 64.100.1.1, timeout is 2 seconds:
Packet sent with a source address of 192.168.2.1
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/12/59 ms

D2# ping 2001:db8:acad:64:100:1::1 source vlan 11
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:ACAD:64:100:1:0:1, timeout is 2
seconds:
Packet sent with a source address of 2001:DB8:ACAD:3000::1
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms

R3# ping 2001:db8:acad:64:100:1::1 source lo 0
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:ACAD:64:100:1:0:1, timeout is 2
seconds:
Packet sent with a source address of 2001:DB8:ACAD:3000::1
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 2/2/4 ms

D2# ping 64.100.1.1 source vlan 11
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 64.100.1.1, timeout is 2 seconds:
Packet sent with a source address of 192.168.3.1
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
```

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